

PASSIVE 70W TESLA T4: DYNAMIC THERMAL THROTTLING VS OCCUPANCY PACING

Comparison of Unconstrained 100% Occupancy (DVFS Thermal Collapse) vs 25% Paced Boost Lock (Lemma 5)

UNCONSTRAINED OCCUPANCY (100% / 1024 THREADS/SM)

1. Maximum Warps Active 32 Warps / SM



2. Dynamic Power Spikes 93.61W (> 70W TDP)



3. Hardware SW_POWER_CAP (0x0004) 72.5% Throttled



4. Frequency Thermal Collapse 1193 MHz Mean (-25%)

PACED OCCUPANCY (25% / 256 THREADS/SM LAUNCH BOUNDS)

1. Paced Launch Bounds __launch_bounds__(128, 2)



2. Thermal Stabilization 50.36W (< 70W Ceiling)



3. Zero Throttling Flag 0.0% Power Throttle



4. Continuous Boost Lock 1590 MHz Locked (1.07x Fast)